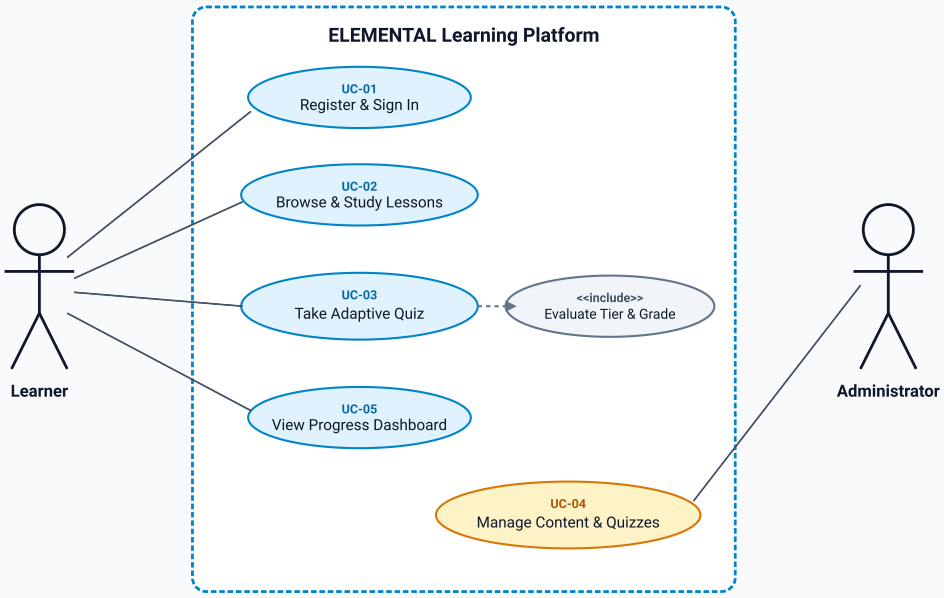
USE CASE MODELING

ELEMENTAL: Use Case Diagram



Explanation: Two primary actors interact with ELEMENTAL. The *Learner* authenticates, studies lessons, takes adaptive quizzes, and monitors overall topic proficiency. The *Administrator* manages levels, topics, lessons, and question banks. UC-03 automatically includes evaluation of performance rules and difficulty tier adjustments.

USE CASE SPECIFICATIONS

Detailed Functional Specifications

UC-01: Register & Sign In	Actor: Learner / Admin
Precondition:	User opens platform on a web or mobile browser.
Main Flow:	1. User submits email/username and password credentials. 2. System validates format and checks cryptographic credentials. 3. System establishes an active user session and returns user profile.
Alternative Flow:	Invalid credentials: System rejects attempt, presents descriptive error message, and allows re-entry.
Postcondition:	Authenticated session state active; user granted role-based access.

UC-02: Browse & Study Lessons	Actor: Learner
Precondition:	Learner is signed in.
Main Flow:	1. Learner selects target TOPIK Level and Topic. 2. System displays structured lesson index. 3. Learner opens selected lesson; system delivers multimedia learning content.
Alternative Flow:	Content unavailable or locked: System displays informative notification guiding user to prerequisite topics.
Postcondition:	Lesson view recorded; learner enabled to start topic practice quiz.

UC-03: Take Adaptive Quiz	Actor: Learner
Precondition:	Learner selects a topic quiz; current difficulty tier evaluated.
Main Flow:	1. System fetches questions matched to learner's current tier. 2. Learner submits completed responses. 3. System grades submission, logs attempt, calculates accuracy score, and evaluates recent performance rules (e.g., ≥80% twice to advance tier).
Alternative Flow:	System detects repeated low scores (<50%): Triggers foundational reinforcement tier and displays targeted review material.
Postcondition:	Quiz attempt saved; learner progress record updated with calculated accuracy and adjusted tier.

UC-04: Manage Content & Quizzes	Actor: Administrator
Precondition:	Administrator is authenticated with elevated permissions.
Main Flow:	1. Admin opens management portal and creates/edits Level, Topic, Lesson, or Question. 2. Admin assigns difficulty tier tags and valid answer keys. 3. System validates content constraints and publishes changes.
Alternative Flow:	Validation fails (missing answer key or incomplete text): System flags errors for correction prior to publishing. Content removal is handled via soft-archiving to preserve student quiz history.
Postcondition:	Educational content updated in central catalog without breaking historical references.

Explanation: Specifications explicitly declare system triggers, normal paths, error handling, and operational postconditions. They ensure functional behavior remains testable, unambiguous, and directly linked to system requirements.